

Biology 30

Unit D: Population and Community Dynamics

Study Guide

1 Unit / Module Objectives

Topic Area	Diploma-Level Objective
Hardy-Weinberg Principle	Apply Hardy-Weinberg equations ($p + q = 1$ and $p^2 + 2pq + q^2 = 1$) to calculate allele and genotype frequencies in populations, and analyze how violations of equilibrium conditions lead to evolutionary change
Factors Causing Gene Pool Change	Evaluate the molecular mechanisms by which natural selection, genetic drift, gene flow, nonrandom mating, bottleneck effects, founder effects, migration, and mutation alter allele frequencies in populations
Species Interactions and Ecological Relationships	Analyze predator-prey dynamics, producer-consumer relationships, and symbiotic interactions (commensalism, mutualism, parasitism) to predict their effects on population structure and community stability
Population Growth Patterns and Regulation	Calculate population parameters using formulas for growth rate ($gr = \Delta N / \Delta t$), per capita growth rate ($cgr = \Delta N / N$), and population density ($Dp = N / A$), and distinguish between exponential (J-shaped) and logistic (S-shaped) growth curves
Reproductive Strategies and Population Characteristics	Compare r-selected and K-selected reproductive strategies in terms of their adaptive advantages under different environmental conditions and their influence on population dynamics
Ecological Succession and Community Development	Distinguish between primary and secondary succession processes, identify pioneer and climax species characteristics, and predict successional outcomes based on environmental disturbance patterns

2 Student Self-Verification Checklist

Topic	Self-Verification Statement
Hardy-Weinberg Calculations	☐ I can calculate allele frequencies using $p + q = 1$ and genotype frequencies using $p^2 + 2pq + q^2 = 1$, and identify when populations deviate from equilibrium conditions

Evolutionary Forces	☐ I can explain how natural selection, genetic drift, gene flow, and mutation change gene pool composition, using examples like antibiotic resistance and herbicide tolerance
Interspecific Relationships	☐ I can classify species interactions as predation, competition, commensalism, mutualism, or parasitism and predict their population-level consequences
Population Mathematics	☐ I can apply population change equations ($\Delta N = [\text{natality} + \text{immigration}] - [\text{mortality} + \text{emigration}]$) and calculate growth rates, per capita growth rates, and population densities
Growth Curve Analysis	☐ I can distinguish exponential from logistic growth patterns, identify carrying capacity and environmental resistance factors, and relate these to r-selected versus K-selected strategies
Succession Dynamics	☐ I can differentiate primary from secondary succession, predict pioneer and climax species characteristics, and analyze how disturbance affects community development

3 Diploma-Level Practice Questions with Fully Worked Solutions

Question	In a population of 1000 individuals, 640 exhibit the dominant phenotype and 360 exhibit the recessive phenotype. Calculate the frequency of the recessive allele (q) and the frequency of heterozygotes in this population, assuming Hardy-Weinberg equilibrium.
Step 1 - Identification	Identify that 360 individuals show recessive phenotype, so they must have genotype qq , representing q^2 in the Hardy-Weinberg equation.
Step 2 - Principle Statement	Calculate $q^2 = 360/1000 = 0.36$, therefore $q = \sqrt{0.36} = 0.6$
Step 3 - Explicit Reasoning	Using $p + q = 1$, calculate $p = 1 - 0.6 = 0.4$. The frequency of heterozygotes is $2pq = 2(0.4)(0.6) = 0.48$. In a population of 1000, this represents 480 heterozygous individuals.
Step 4 - Validation	Verify by checking that $p^2 + 2pq + q^2 = (0.4)^2 + 2(0.4)(0.6) + (0.6)^2 = 0.16 + 0.48 + 0.36 = 1.0$
Step 5 - Final Answer	The recessive allele frequency (q) is 0.6 or 60%, and the frequency of heterozygotes is 0.48 or 48% (480 individuals).
Common Error	Students often confuse phenotype frequencies with genotype frequencies, incorrectly using the 640 dominant phenotype individuals

	as p^2 instead of recognizing that dominant phenotype includes both homozygous dominant (p^2) and heterozygous ($2pq$) individuals.
Exam Strategy	Always start Hardy-Weinberg problems by identifying the recessive phenotype first, since these individuals must be homozygous recessive (q^2), giving you a direct path to calculate q before finding p .
Question	A small population of endangered birds experiences a bottleneck event, reducing from 10,000 to 50 individuals. Explain the genetic consequences and predict how this affects the population's evolutionary potential.
Step 1 - Identification	Define bottleneck effect as a dramatic reduction in population size that causes random loss of alleles and reduces genetic diversity.
Step 2 - Principle Statement	Calculate the severity: population reduced by 99.5% ($9,950/10,000$), representing an extreme bottleneck with severe genetic drift.
Step 3 - Explicit Reasoning	Genetic consequences include random allele loss regardless of adaptive value, reduced heterozygosity, increased homozygosity, and potential fixation of harmful alleles. The effective population size (N_e) becomes much smaller than census population, amplifying drift effects. Rare alleles (frequency <0.02) have high probability of being lost completely.
Step 4 - Validation	Evolutionary potential decreases dramatically due to reduced raw material for natural selection, increased inbreeding depression, and loss of adaptive alleles that might be crucial for future environmental changes.
Step 5 - Final Answer	The bottleneck severely reduces genetic diversity, limits evolutionary potential through genetic drift overwhelming natural selection, and increases extinction risk due to reduced adaptive capacity and potential inbreeding depression.
Common Error	Students often focus only on the immediate population size reduction without recognizing that the long-term genetic effects persist even if population size recovers, creating a permanent loss of genetic diversity.
Exam Strategy	For bottleneck questions, emphasize the irreversible nature of allele loss and connect small population size to genetic drift overpowering natural selection, making evolution more random than adaptive.
Question	Describe the ecological relationship between cleaner fish and their host fish, and predict what would happen to both populations if the cleaner fish were removed from the ecosystem.
Step 1 - Identification	Identify this as a mutualistic symbiotic relationship where cleaner fish (like wrasses) remove parasites and dead tissue from host fish.

Step 2 - Principle Statement	Analyze benefits: cleaner fish receive food (parasites, dead tissue, mucus), while host fish gain parasite removal, wound cleaning, and improved health.
Step 3 - Explicit Reasoning	Both species have evolved specific adaptations for this interaction: cleaner fish have specialized mouth structures and behavior patterns, while host fish exhibit cleaning station behavior and submission postures. The relationship involves specific recognition signals and territories (cleaning stations) where the interaction occurs safely.
Step 4 - Validation	Predict consequences of cleaner fish removal: host fish populations would experience increased parasite loads, reduced health, potential disease outbreaks, and possible population decline.
Step 5 - Final Answer	This mutualistic relationship where cleaner fish removal would cause increased parasite burden and disease in host fish, potentially leading to population crashes, while cleaner fish would lose their primary food source and likely face local extinction.
Common Error	Students often misclassify this as parasitism because one fish is on another, failing to recognize that both species benefit and that the cleaner fish provides a service rather than causing harm.
Exam Strategy	For symbiosis questions, always identify what each organism gains or loses, then classify based on the net effect (+/+, +/0, +/-) rather than just the physical appearance of the interaction.
Question	A closed population starts with 100 individuals. During one year, 25 offspring are born and 15 individuals die. Calculate the population growth rate (gr) and per capita growth rate (cgr) for this population.
Step 1 - Identification	Identify given values: initial population (N) = 100, natality = 25, mortality = 15, time period (Δt) = 1 year.
Step 2 - Principle Statement	Calculate population change using $\Delta N = (\text{natality} + \text{immigration}) - (\text{mortality} + \text{emigration})$. Since closed population, immigration = emigration = 0.
Step 3 - Explicit Reasoning	$\Delta N = (25 + 0) - (15 + 0) = 10$ individuals. Growth rate (gr) = $\Delta N / \Delta t = 10 / 1 = 10$ individuals per year. Per capita growth rate (cgr) = $\Delta N / N = 10 / 100 = 0.1$ per individual per year.
Step 4 - Validation	Verify units and biological meaning: gr shows absolute change in population size over time, while cgr shows proportional change per existing individual.

Step 5 - Final Answer	The population growth rate (gr) is 10 individuals per year, and the per capita growth rate (cgr) is 0.1 per individual per year or 10% annual growth rate.
Common Error	Students often confuse growth rate (gr) with per capita growth rate (cgr), using the wrong denominator in calculations or forgetting to specify that cgr represents growth per existing individual.
Exam Strategy	Remember that gr uses time in denominator ($\Delta N/\Delta t$) while cgr uses population size in denominator ($\Delta N/N$), and cgr can be converted to percentage by multiplying by 100.
Question	Compare the characteristics of r-selected and K-selected species, and explain why r-selected species are more likely to be found in disturbed habitats while K-selected species dominate stable environments.
Step 1 - Identification	Define r-selected species as those adapted for rapid population growth in unstable environments, and K-selected species as those adapted for competitive ability near carrying capacity.
Step 2 - Principle Statement	List contrasting characteristics: r-selected have high reproductive rate, small body size, short lifespan, early maturity, many small offspring, little parental care; K-selected have low reproductive rate, large body size, long lifespan, delayed maturity, few large offspring, extensive parental care.
Step 3 - Explicit Reasoning	In disturbed habitats, resources become temporarily abundant and competition is reduced, favoring rapid reproduction and colonization ability. R-selected species can quickly exploit these conditions before competition intensifies. In stable environments near carrying capacity (K), competition for limited resources favors efficient resource use, competitive ability, and investment in offspring quality over quantity.
Step 4 - Validation	Environmental unpredictability favors 'bet-hedging' through many offspring, while environmental stability favors investment in competitive superiority and offspring survival.
Step 5 - Final Answer	R-selected species (high reproductive rate, small size, early maturity) thrive in disturbed habitats due to rapid colonization ability, while K-selected species (low reproductive rate, large size, competitive ability) dominate stable environments through superior competitive ability near carrying capacity.
Common Error	Students often memorize the characteristics without understanding the underlying evolutionary logic that links reproductive strategy to environmental predictability and competitive intensity.

Exam Strategy	Connect r and K strategies to their namesake parameters in logistic growth: r-selected maximize intrinsic growth rate, K-selected maximize competitive ability near carrying capacity.
Question	Distinguish between primary and secondary succession, providing specific examples of pioneer species and explaining why climax communities have greater species diversity than early successional stages.
Step 1 - Identification	Define primary succession as occurring on previously uncolonized surfaces (volcanic rock, glacial till, new sand dunes), while secondary succession occurs after disturbance on previously vegetated areas (forest fires, agricultural abandonment).
Step 2 - Principle Statement	Identify pioneer species examples: primary succession features lichens, mosses, and hardy grasses that can establish on bare rock or poor soil; secondary succession features fast-growing herbs, shrubs, and early successional trees like aspen or birch.
Step 3 - Explicit Reasoning	Succession proceeds through predictable stages as pioneer species modify environmental conditions (soil formation, nutrient accumulation, microclimate changes), creating conditions suitable for later successional species. Each stage facilitates the next through facilitation, inhibition, or neutral interactions. Species diversity increases as habitat heterogeneity increases and niche specialization develops.
Step 4 - Validation	Climax communities represent stable endpoints with complex trophic structures, diverse microhabitats, and specialized species interactions that support higher species richness.
Step 5 - Final Answer	Primary succession starts on sterile substrates with pioneer species like lichens, while secondary succession begins on disturbed but previously vegetated areas with pioneers like grasses and shrubs. Climax communities have greater diversity due to increased habitat complexity, niche specialization, and stable environmental conditions.
Common Error	Students often confuse the starting conditions, thinking secondary succession always takes longer than primary succession, when actually secondary succession is usually faster due to existing soil and seed banks.
Exam Strategy	Remember 'primary = pristine/bare' and 'secondary = soil still there' to distinguish succession types, then think about time scales and which pioneer species could actually survive the initial harsh conditions.

Question	Explain how antibiotic-resistant bacteria exemplify natural selection in action, describing the molecular mechanisms involved and predicting the consequences of antibiotic overuse for human health.
Step 1 - Identification	Identify antibiotic resistance as a clear example of directional natural selection where antibiotic pressure selects for resistant bacterial genotypes.
Step 2 - Principle Statement	Describe molecular mechanisms: resistance genes can arise through random mutations (e.g., altered target proteins, efflux pumps) or horizontal gene transfer (plasmids carrying resistance genes between bacteria).
Step 3 - Explicit Reasoning	Natural selection process: in presence of antibiotics, susceptible bacteria die while resistant bacteria survive and reproduce. Resistance alleles increase in frequency each generation. Mathematical relationship follows Hardy-Weinberg deviations where selection coefficient (s) reduces fitness of susceptible genotypes, causing allele frequency changes over time according to $\Delta p = sp(1-p)/[1-sp]$.
Step 4 - Validation	Antibiotic overuse increases selection pressure, accelerates resistance evolution, and can lead to multi-drug resistant 'superbugs' through accumulation of multiple resistance genes.
Step 5 - Final Answer	Antibiotic-resistant bacteria demonstrate natural selection through survival and reproduction of resistant genotypes under antibiotic pressure, involving molecular mechanisms like target modification and gene transfer. Overuse accelerates resistance evolution, creating serious public health threats.
Common Error	Students often think bacteria 'become resistant' in response to antibiotics, missing that resistance mutations exist randomly before antibiotic exposure and selection simply increases their frequency.
Exam Strategy	Emphasize that selection acts on existing genetic variation, not by creating new mutations in response to environmental pressure - the resistant bacteria were already there in low frequencies.
Question	A predator population of 50 wolves supports a prey population of 2000 deer in a 400 km ² area. Calculate the population density for both species and explain how changes in wolf density would affect deer population dynamics through predator-prey interactions.
Step 1 - Identification	Apply population density formula $D_p = N/A$ where N is population size and A is area.

Step 2 - Principle Statement	Calculate wolf density: $D_p(\text{wolves}) = 50 \text{ wolves} / 400 \text{ km}^2 = 0.125 \text{ wolves/km}^2$. Calculate deer density: $D_p(\text{deer}) = 2000 \text{ deer} / 400 \text{ km}^2 = 5 \text{ deer/km}^2$.
Step 3 - Explicit Reasoning	Predator-prey dynamics follow cyclical patterns where predator density affects prey mortality rates, and prey density affects predator natality and survival. Increased wolf density would increase deer mortality through higher predation rates, potentially reducing deer population size. However, as deer density decreases, intraspecific competition among deer decreases, potentially improving survival and reproduction of remaining deer. Wolf population would eventually decline due to reduced prey availability, creating oscillating population cycles.
Step 4 - Validation	Additional factors include deer carrying capacity, alternative prey availability for wolves, and environmental factors affecting both species.
Step 5 - Final Answer	Wolf density is $0.125 \text{ wolves/km}^2$ and deer density is 5 deer/km^2 . Increased wolf density would initially reduce deer populations through higher predation pressure, but predator-prey dynamics create cyclical population fluctuations as wolf numbers eventually decline when prey becomes scarce.
Common Error	Students often predict simple linear relationships between predator and prey numbers without recognizing the time lags and cyclical nature of predator-prey dynamics that create population oscillations.
Exam Strategy	For predator-prey questions, always consider time delays and feedback loops: more predators reduce prey, which eventually reduces predator food supply, leading to predator decline and prey recovery.